CSE 5710 - Data Mining for Data Science

Numeric Attributes

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Objectives

Objectives:

- Discuss the statistical methods for exploratory data analysis of *numeric attributes* in a dataset.
- Focus on identifying the following metrics and techniques for measuring the following during data exploration:
  - Measures of *central tendency*
  - Measures of *dispersion*
  - Measures of *linear dependence* or *association* between attributes
- Identify the theoretical measures for the population, and techniques for estimating them using our dataset (which we view as an extract from the larger population).
- Analyze the relationship between the *probabilistic perspective* and the *geometric perspective* on these metrics.

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Univariate Analysis

Definition

Univariate Analysis is the process of analyzing the characteristics of a *single* attribute, typically using a statistical sample of random independent observations of that attribute.

Univariate Analysis

The data matrix $\boldsymbol{D}$ is an $n \times 1$ matrix,

$$\boldsymbol{D} = \begin{pmatrix} X \\ \hline x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

where X is the numeric attribute of interest, with $x_i \in \mathbb{R}$.

Univariate Analysis

$$\boldsymbol{D} = \begin{pmatrix} X \\ \hline x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

Probabilistic Perspective:

- X is assumed to be a random variable with a particular probability distribution.
- x_i's constitute an independent, identically distributed (IID) random sample of n observations of the random variable X.

Univariate Analysis

$$\boldsymbol{D} = \begin{pmatrix} X \\ \hline x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

Geometric (Vector) Perspective:

- Treat the sample as a *single n-dimensional vector* $\Rightarrow X \in \mathbb{R}^n$.

Theoretical $f(x)$, $F(x)$

For random variable, X, we have the following functions that define the distribution of X:

- $f(x)$: defines the $p(X = x)\ \forall x \in \mathbb{R}$.
- $F(x)$: defines the $p(X <= x)\ \forall x \in \mathbb{R}$.

However, these are generally unknown.

So, we commonly estimate these functions using the techniques in the following slides . . .

Empirical Probability Mass Function

Definition

The **empirical probability mass function (PMF)** of X is given as follows:

$$\hat{f}(x) = P(X = x) = \frac{1}{n} \sum_{i=1}^{n} I(x_i = x)$$

where the indicator variable I takes on the value 1 when its argument is true, and 0 otherwise, i.e.,

$$I(x_i = x) = \begin{cases} 1 & \text{if } x_i = x \\ 0 & \text{if } x \neq x \end{cases}$$

The empirical PMF puts a probability mass of $\frac{1}{n}$ at each point x_i.

Empirical Cumulative Distribution Function

Definition

The **empirical cumulative distribution function (CDF)** of X is given as follows:

$$\hat{F}(x) = \frac{1}{n} \sum_{i=1}^{n} I(x_i \leq x)$$

where

$$I(x_i \leq x) = \begin{cases} 1 & \text{if } x_i \leq x \\ 0 & \text{otherwise} \end{cases}$$

Empirical Inverse Cumulative Distribution Function

Definition

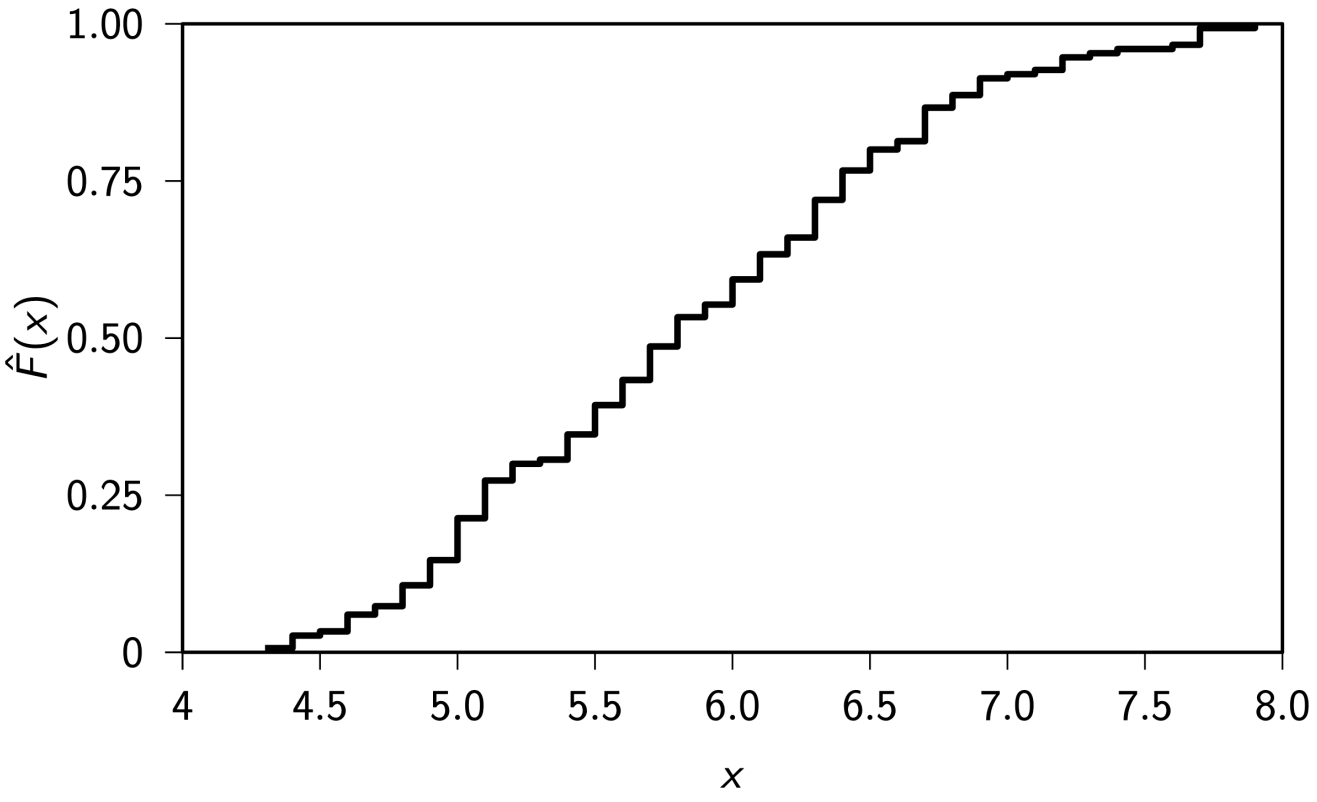
The **inverse cumulative distribution function** or **quantile function** for X is defined as follows:

$$F^{-1}(q) = \min\{x \mid \hat{F}(x) \geq q\} \qquad \text{for } q \in [0, 1]$$

The inverse CDF gives the least value of X, for which q fraction of the values are higher, and $1 - q$ fraction of the values are lower.

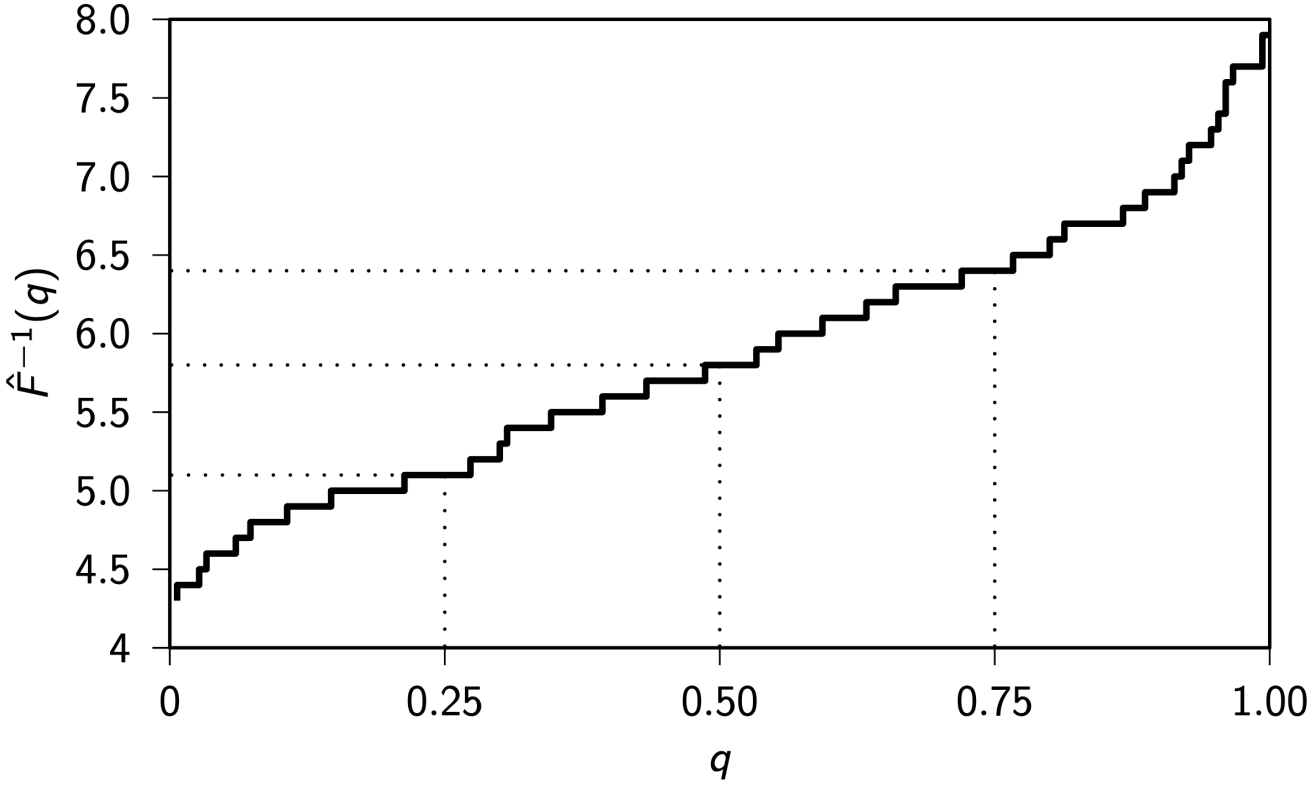
Example: Iris Dataset Extract

Empirical CDF for sepal length



Example: Iris Dataset Extract

Empirical Inverse CDF for sepal length



The median is 5.8, since

$$\hat{F}(5.8) = 0.5 \text{ or } 5.8 = \hat{F}^{-1}(0.5)$$

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Measures of Central Tendency

The following slides give measures of:

- concentration of the probability mass
- "middle" values
- most frequent values
- others...

Mean

Definition

The **mean** or **expected value** of a random variable X is the arithmetic weighted average of the values of X, weighted by their probability of occurrence.

It provides a one-number summary of the *location* or *central tendency* for the distribution of X.

Mean: Discrete Case

Definition

Discrete Case:

$$\mu = E[X] = \sum_x x \cdot f(x)$$

where $f(x)$ is the probability mass function of X.

Mean: Continuous Case

Definition

Continuous Case:

$$\mu = E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

where $f(x)$ is the probability density function of X.

Estimator for Mean: Sample Mean

Definition

The **sample mean** is a statistic, that is, a function $\hat{\mu} : \{x_1, x_2, \ldots, x_n\} \to \mathbb{R}$, defined as the average value of x_i's:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

The sample means serves as an estimator for the unknown mean value μ of X.

Estimator for Mean: Sample Mean

We can also arrive at this formula by substituting empirical probabiliy mass function $\hat{f}(x)$ for the theoretical PMF, $f(x)$ in the formula for $E(X)$ for the discrete case:

$$\hat{\mu} = \sum_x x \cdot \hat{f}(x) = \sum_x x \left(\frac{1}{n} \sum_{i=1}^{n} I(x_i = x) \right) = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Unbiased Estimator

Definition

An estimator $\hat{\theta}$ is called an **unbiased estimator** for parameter θ if $E[\hat{\theta}] = \theta$ for every possible value of θ.

Sample Mean: Unbiased Estimator

Consider the following for the sample mean $\hat{\mu}$:

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$$E[\hat{\mu}] = E\left[\frac{1}{n}\sum_{i=1}^{n} x_i\right]$$

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Sample Mean: Unbiased Estimator

Consider the following for the sample mean $\hat{\mu}$:

$$E[\hat{\mu}] = E\left[\frac{1}{n}\sum_{i=1}^{n} x_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[x_i] = \frac{1}{n}\sum_{i=1}^{n} \mu$$

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Sample Mean: Unbiased Estimator

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Thus, $\hat{\mu}$ is an unbiased estimator for the population mean μ.

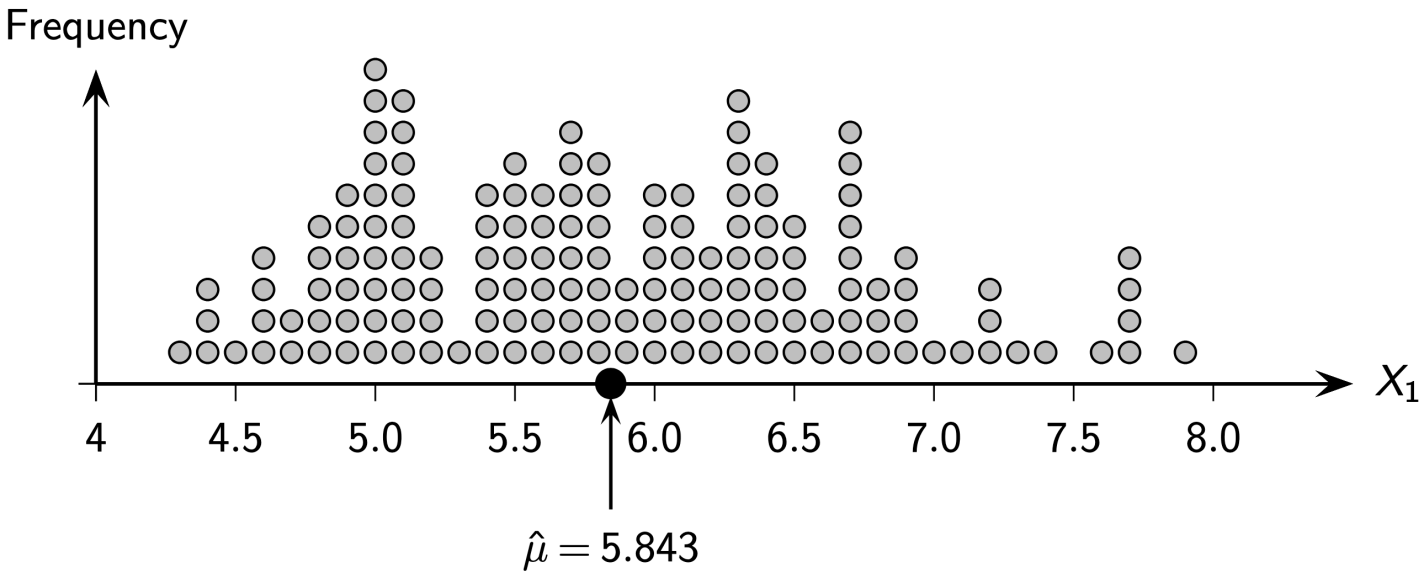
Sample Mean: Not Robust

Definition

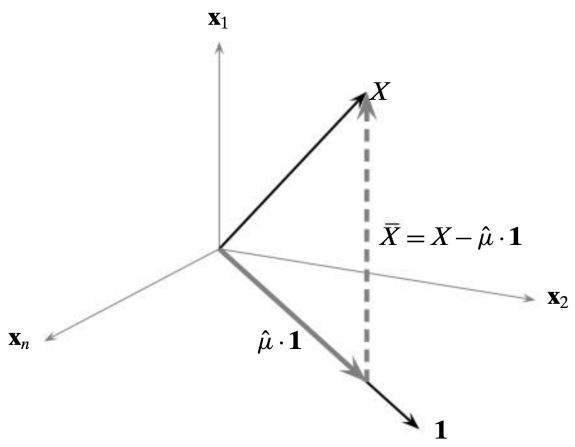
We say that a statistic is **robust** if it is not affected by extreme values (such as outliers) in the data.

The sample mean is not robust because a single large value can skew the average.

Example: Iris Dataset Extract - Sepal Length

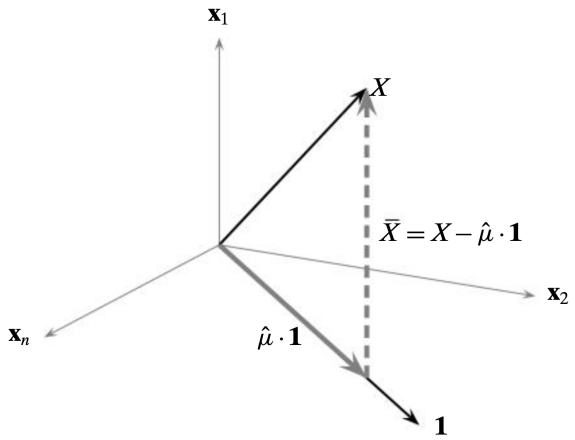


Geometric Interpretation of the Sample Mean



Geometric Perspective:

- Consider X as a vector in n-dimensional space, i.e., $X = (x_1, x_2, \cdots, x_n)^T \in \mathbb{R}^n$ where n is the sample size.
- Let $\mathbf{1} = (1, 1, \cdots, 1)^T \in \mathbb{R}^n$ be the *ones vector* i.e., the vector all of whose elements are 1.



Geometric Perspective:

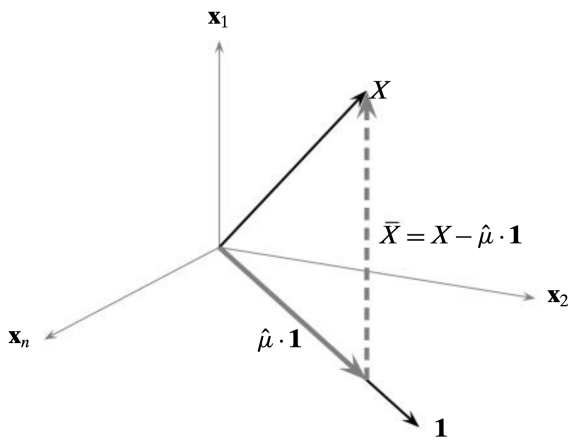
- Consider the projection of X onto the vector, $\mathbf{1}$.
- In $\mathbb{R}^n$, the projection, $\boldsymbol{p}$ of any vector, $\boldsymbol{b}$ onto vector $\boldsymbol{a}$ is given by:

$$\boldsymbol{p} = \left(\frac{\boldsymbol{a}^T \boldsymbol{b}}{\boldsymbol{a}^T \boldsymbol{a}} \right) \boldsymbol{a}$$

- In this case, we have:

$$\boldsymbol{p} = \left(\frac{X^T \mathbf{1}}{\mathbf{1}^T \mathbf{1}} \right) \mathbf{1} = \left(\frac{\sum_{i=1}^{n} x_i}{n} \right) \mathbf{1} = \hat{\mu} \cdot \mathbf{1}$$

Note: $\hat{\mu}$ is a scalar, $\therefore \hat{\mu} \cdot \mathbf{1}$ is *not* a dot-product



Geometric Perspective:

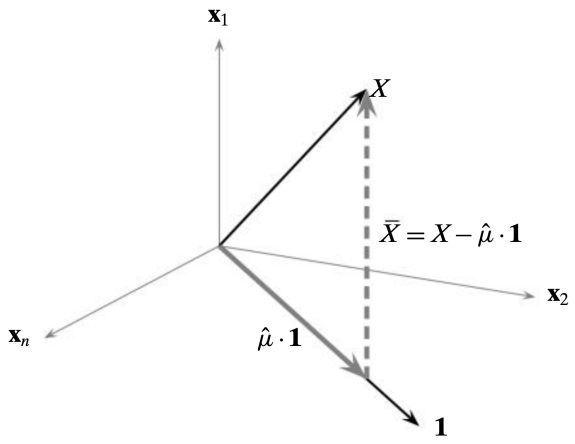
- Thus, we have :

$$\boldsymbol{p} = \hat{\mu} \cdot \mathbf{1}$$

  This means that $\hat{\mu}$ is the scalar projection of X onto the vector $\mathbf{1}$.

- That is,

$$\hat{\mu} = \text{proj}_{\mathbf{1}}(X) = \left(\frac{X^T \mathbf{1}}{\mathbf{1}^T \mathbf{1}} \right)$$

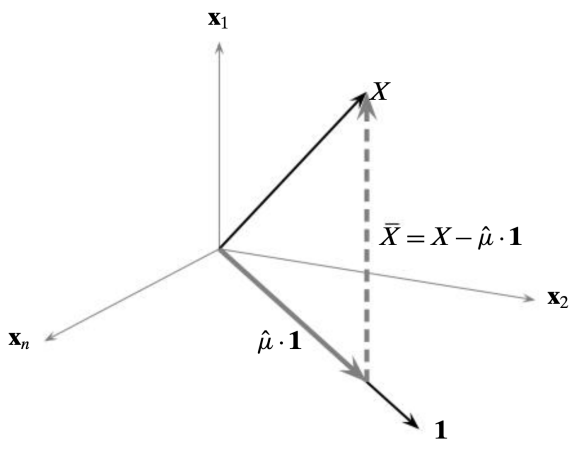


Geometric Perspective:

- The sample mean can be used to center the attribute X.
- Define:

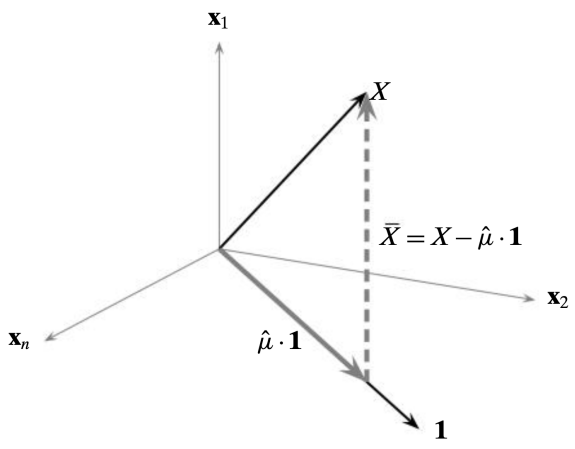
$$\bar{X} = X - \hat{\mu} \cdot \mathbf{1} = \begin{pmatrix} x_1 - \hat{\mu} \\ x_2 - \hat{\mu} \\ \vdots \\ x_n - \hat{\mu} \end{pmatrix}$$

- We can see from the diagram that $\hat{\mu} \cdot \mathbf{1}$ is the parallel component and $\bar{X}$ is the orthogonal component of the projection of X onto $\mathbf{1}$.



Geometric Perspective:

- Note that $\mathbf{1}$ and $\bar{X}$ are orthogonal:



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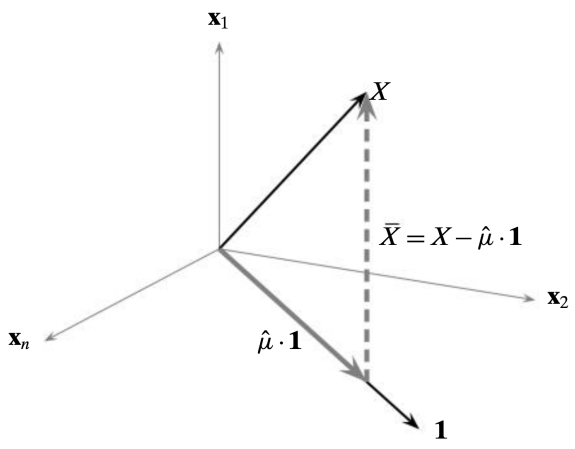
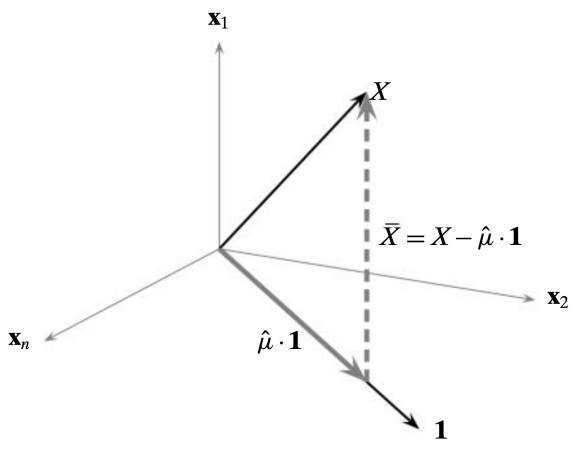


Geometric Perspective:

- Note that $\mathbf{1}$ and $\bar{X}$ are orthogonal:

$$\mathbf{1}^T \bar{X} = \mathbf{1}^T (X - \hat{\mu} \cdot \mathbf{1})$$



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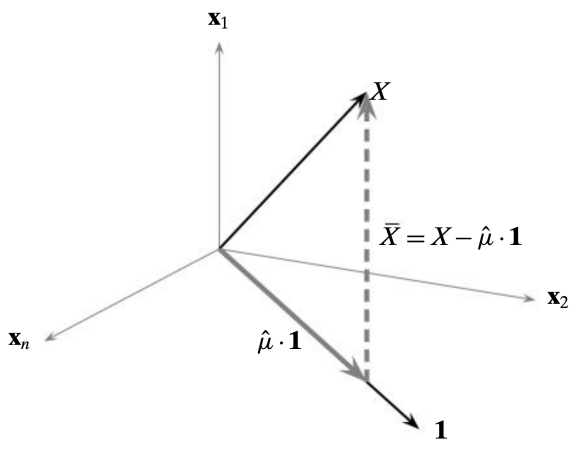
$$\begin{aligned}\mathbf{1}^T \bar{X} &= \mathbf{1}^T (X - \hat{\mu} \cdot \mathbf{1}) \\ &= \mathbf{1}^T \left(X - \left(\frac{X^T \mathbf{1}}{\mathbf{1}^T \mathbf{1}} \right) \cdot \mathbf{1} \right)\end{aligned}$$



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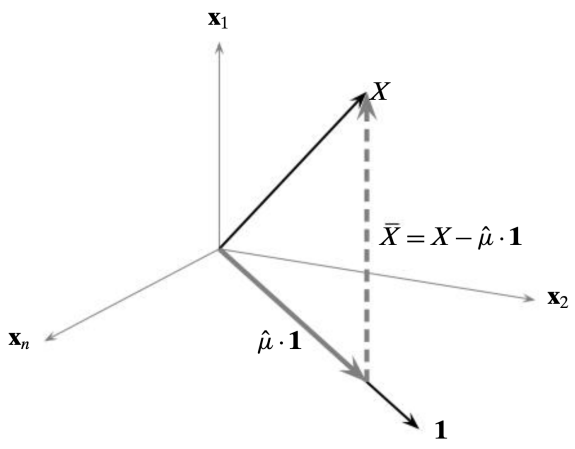
\mathbf{1}^T \bar{X} &= \mathbf{1}^T (X - \hat{\mu} \cdot \mathbf{1}) \\
&= \mathbf{1}^T \left(X - \left(\frac{X^T \mathbf{1}}{\mathbf{1}^T \mathbf{1}} \right) \cdot \mathbf{1} \right) \\
&= \mathbf{1}^T X - \left(\frac{X^T \mathbf{1}}{\mathbf{1}^T \mathbf{1}} \right) \cdot \mathbf{1}^T \mathbf{1}
\end{aligned}$$



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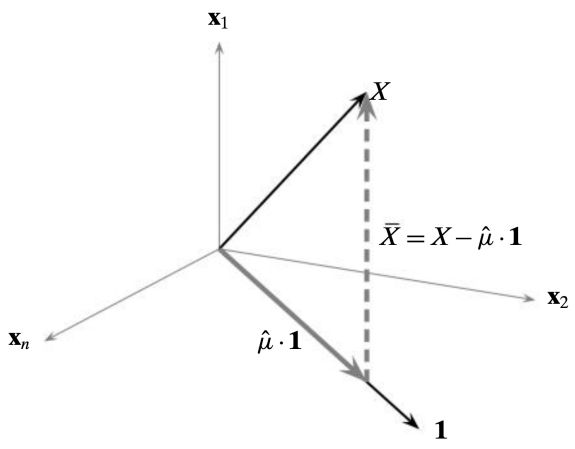

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&= \mathbf{1}^T X - \mathbf{1}^T X \\
&= 0
\end{aligned}
$$

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- We can compute its mean value and variance as follows:

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\begin{aligned}
E[\hat{\mu}] &= \mu \\
\mathrm{var}(\hat{\mu}) &= E\left[(\hat{\mu} - \mu)^2\right] \qquad \text{by definition of variance}
\end{aligned}
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&= \frac{1}{n^2}\,\mathrm{var}\left(\sum_{i=1}^{n} x_i\right) \\
&= \frac{1}{n^2}\left(\sum_{i=1}^{n} \mathrm{var}(x_i)\right) \qquad \text{due to independence}
\end{aligned}
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&= \frac{1}{n^2}\operatorname{var}\left(\sum_{i=1}^{n} x_i\right) \\
&= \frac{1}{n^2}\left(\sum_{i=1}^{n}\operatorname{var}(x_i)\right) \qquad \text{due to independence} \\
&= \frac{1}{n^2}\left(n\operatorname{var}(x_i)\right)
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&= \frac{1}{n^2}\left(n\operatorname{var}(x_i)\right) \\
&= \frac{\operatorname{var}(x_i)}{n}
\end{aligned}
$$

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$$
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&= \frac{1}{n^2}\left(\sum_{i=1}^{n}\operatorname{var}(x_i)\right) \qquad \text{due to independence} \\
&= \frac{1}{n^2}\left(n\operatorname{var}(x_i)\right) \\
&= \frac{\operatorname{var}(x_i)}{n} = \frac{\sigma^2}{n} \qquad \text{since } x_i \text{ has distribution of } X \Rightarrow \operatorname{var}(x_i) = \sigma^2
\end{aligned}
$$

Median

Definition

The **median** of a random variable is defined as the value m such that

$$P(X \leq m) \geq \frac{1}{2} \text{ and } P(X \geq m) \geq \frac{1}{2}$$

- The median m is the "middle-most" value; half of the values of X are less and half of the values of X are more than m.
- In terms of the (inverse) cumulative distribution function, the median is the value m for which

$$F(m) = 0.5 \text{ or } m = F^{-1}(0.5)$$

- The median is robust, as it is not affected very much by extreme values.

Sample Median

Definition

The **sample median** is given by the following:

$$\hat{F}(m) = 0.5 \text{ or } m = \hat{F}^{-1}(0.5)$$

Mode

Definition

The **mode** of a random variable X is the value at which the probability mass function or probability density function attains its maximum value, depending on whether X is discrete or continuous, respectively.

Sample Mode

Definition

The **sample mode** is a value for which the empirical probability mass function attains its maximum, given as

$$\text{mode}(X) = \arg\max_{x} \hat{f}(x)$$

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   - Fundamentals
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   - Measures of Dispersion
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3. Bivariate Analysis
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   - Fundamentals
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Range

Definition

The **value range** or simply **range** of a random variable X is the difference between the maximum and minimum values of X, given as follows:

$$r = \max\{X\} - \min\{X\}$$

Sample Range

Definition

The **sample range** is a statistic, given as follows:

$$\hat{r} = \max_{i=1}^{n}\{x_i\} - \min_{i=1}^{n}\{x_i\}$$

- Range is sensitive to extreme values, and thus is not robust.

Interquartile Range

Definition

A more robust measure of the dispersion of X is the **interquartile range (IQR)**, which is defined as follows:

$$IQR = F^{-1}(0.75) - F^{-1}(0.25)$$

Sample IQR

Definition

The **sample IQR** is given as:

$$\widehat{IQR} = \hat{F}^{-1}(0.75) - \hat{F}^{-1}(0.25)$$

Variance and Standard Deviation

Definition

The **variance** of a random variable X provides a measure of how much the values of X deviate from the mean or expected value of X, given as follows:

$$\sigma^2 = \mathrm{var}(X) = E\big[(X-\mu)^2\big] = \begin{cases} \displaystyle\sum_x (x-\mu)^2\, f(x) & \text{if } X \text{ discrete} \\ \displaystyle\int_{-\infty}^{\infty} (x-\mu)^2\, f(x)\, dx & X \text{ continuous} \end{cases}$$

The **standard deviation** σ, is the positive square root of the variance, σ^2.

Sample Variance

Definition

The **sample variance** is defined as

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\mu})^2$$

and the **sample standard deviation** is given by the following:

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\mu})^2}$$

Geometric Interpretation of Sample Variance

Geometric Perspective for Sample Variance:

- Consider X as a vector in n-dimensional space where n is the sample size, i.e., $X = (x_1, x_2, \cdots, x_n) \in \mathbb{R}^n$.
- Let $\bar{X}$ denote the centered sample

$$\bar{X} = X - \mathbf{1} \cdot \hat{\mu} = \begin{pmatrix} x_1 - \hat{\mu} \\ x_2 - \hat{\mu} \\ \vdots \\ x_n - \hat{\mu} \end{pmatrix}$$

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- Therefore, the sample variance is squared magnitude of the centered attribute vector, normalized by the sample size.

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- Using this identity, we have the following:

$$\sum_{i=1}^{n}(x_i - \mu)^2 = \sum_{i=1}^{n}(x_i - \hat{\mu})^2 + n(\hat{\mu} - \mu)^2$$

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Now, $E\left(\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2\right) = \frac{1}{n}nE\left[(x_i - \mu)^2\right] = E\left[(x_i - \mu)^2\right] = \sigma^2$

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&= \sigma^2 - E\left[(\hat{\mu} - \mu)^2\right]
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&= \sigma^2 - E\left[(\hat{\mu} - \mu)^2\right] \\
&= \sigma^2 - \mathrm{var}(\hat{\mu}) \\
&= \sigma^2 - \frac{\sigma^2}{n} \\
E(\hat{\sigma}^2) &= \left(\frac{n-1}{n}\right)\sigma^2
\end{aligned}
$$

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Bivariate Analysis

Definition

Bivariate Analysis is the process of analyzing the characteristics of a *two* attributes, typically using a statistical sample of n *pairs* of random independent observations where each pair consists of an observation of the first attribute and one of the second attribute.

Bivariate Analysis

The data matrix $\boldsymbol{D}$ is an $n \times 2$ matrix,

$$\boldsymbol{D} = \begin{pmatrix} X_1 & X_2 \\ \hline x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}$$

where

$X_1 :=$ first attribute random variable (RV), with $x_{i1} \in \mathbb{R}$ for $i = 1, 2, \ldots n$
$X_2 :=$ second attribute RV, with paired $x_{i2} \in \mathbb{R}$ for $i = 1, 2, \ldots n$

Bivariate Analysis

$$\boldsymbol{D} = \begin{pmatrix} X_1 & X_2 \\ \hline x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}$$

Probabilistic Perspective:

- $\boldsymbol{X} = (X_1, X_2)^T$ is a bivariate vector random variable and the points $\boldsymbol{x}_i$ $(1 \leq i \leq n)$ constitute a random sample of n IID observations of $\boldsymbol{X}$.

Bivariate Analysis

$$\boldsymbol{D} = \begin{pmatrix} X_1 & X_2 \\ \hline x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \\ x_{n1} & x_{n2} \end{pmatrix}$$

Geometric (Vector) Perspective:

- $\boldsymbol{D}$ comprises n points or vectors in 2-dimensional space

$$\boldsymbol{x}_i = (x_{i1}, x_{i2})^T \in \mathbb{R}^2$$

*** OR ***

- $\boldsymbol{D}$ can also be viewed as two points or vectors in an n-dimensional space:

$$X_1 = (x_{11}, x_{21}, \ldots, x_{n1})^T$$
$$X_2 = (x_{12}, x_{22}, \ldots, x_{n2})^T$$

Empirical Joint Probability Mass Function

Definition

The **empirical joint probability mass function (PMF)** of X is given as follows:

$$\hat{f}(\boldsymbol{x}) = P(\boldsymbol{X} = \boldsymbol{x}) = \frac{1}{n} \sum_{i=1}^{n} I(\boldsymbol{x}_i = \boldsymbol{x})$$

$$\hat{f}(x_1, x_2) = P(X_1 = x_1, X_2 = x_2) = \frac{1}{n} \sum_{i=1}^{n} I(x_{i1} = x_1, x_{i2} = x_2)$$

where $\boldsymbol{x} = (x_1, x_2)^T$ and I is an indicator variable that takes on the value 1 only when its argument is true:

$$I(\boldsymbol{x}_i = \boldsymbol{x}) = \begin{cases} 1 & \text{if } x_{i1} = x_1 \text{ and } x_{i2} = x_2 \\ 0 & \text{if otherwise} \end{cases}$$

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Bivariate Mean

Definition

The **bivariate mean vector** is defined as the expected value of the vector random variable $\boldsymbol{X}$:

$$\boldsymbol{\mu} = E[\boldsymbol{X}] = E\left[\begin{pmatrix} X_1 \\ X_2 \end{pmatrix}\right] = \begin{pmatrix} E[X_1] \\ E[X_2] \end{pmatrix} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

That is, $\boldsymbol{\mu}$ is just the vector of expected values of X_1 and X_2.

Sample Bivariate Mean

Definition

The **sample bivariate mean vector** is given as

$$\hat{\boldsymbol{\mu}} = \sum_{\boldsymbol{x}} \boldsymbol{x} \hat{f}(\boldsymbol{x}) = \sum_{\boldsymbol{x}} \boldsymbol{x} \left(\frac{1}{n} \sum_{i=1}^{n} I(\boldsymbol{x}_i = \boldsymbol{x}) \right) = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{x}_i$$

Attribute Variances and Total Variance

The variances of the individual attributes are given by σ_1^2 for X_1 and σ_2^2 for X_2.

Definition

The **total variance** is given by:

$$\text{total variance } = \sigma_1^2 + \sigma_2^2$$

Sample Total Variance

The sample variances of the individual attributes can be computed, $\hat{\sigma}_1^2$ for X_1 and $\hat{\sigma}_2^2$ for X_2.

Definition

The **sample total variance** is given by:

$$var(\boldsymbol{D}) = \hat{\sigma}_1^2 + \hat{\sigma}_2^2$$

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Covariance

Definition

The **covariance** between two attributes X_1 and X_2 provides a measure of the association or linear dependence between them, and is defined as follows:

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$$\sigma_{12} = E[(X_1 - \mu_1)(X_2 - \mu_2)]$$

Covariance

Note, we can derive the following formula based on this definition:

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&= E[X_1X_2] - \mu_1\mu_2 - \mu_1\mu_2 + \mu_1\mu_2 \\
&= E[X_1X_2] - \mu_1\mu_2
\end{aligned}$$

$$\boxed{\sigma_{12} = E[X_1X_2] - E[X_1]E[X_2]}$$

Sample Covariance

Definition

The **sample covariance** between X_1 and X_2 is given as

$$\hat{\sigma}_{12} = \frac{1}{n} \sum_{i=1}^{n} (x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2)$$

Correlation

Definition

The **correlation** between variables X_1 and X_2, also known as the **standardized covariance**, obtained by normalizing the covariance with the standard deviation of each variable, is given as follows:

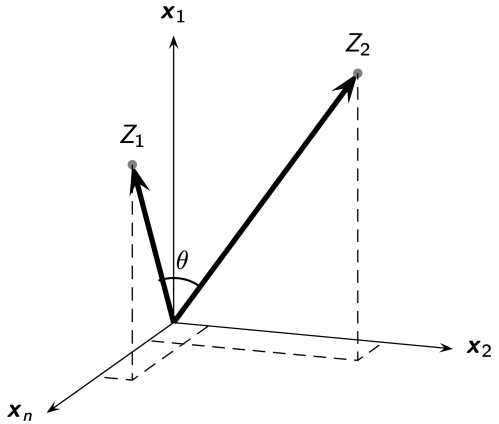
$$\rho_{12} = \frac{\sigma_{12}}{\sigma_1 \sigma_2} = \frac{\sigma_{12}}{\sqrt{\sigma_1^2 \sigma_2^2}}$$

Sample Correlation

Definition

The **sample correlation** for attributes X_1 and X_2 is given as follows:

$$\hat{\rho}_{12} = \frac{\hat{\sigma}_{12}}{\hat{\sigma}_1 \hat{\sigma}_2} = \frac{\sum_{i=1}^{n}(x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2)}{\sqrt{\sum_{i=1}^{n}(x_{i1} - \hat{\mu}_1)^2 \sum_{i=1}^{n}(x_{i2} - \hat{\mu}_2)^2}}$$



Geometric Interpretation of Sample Covariance:

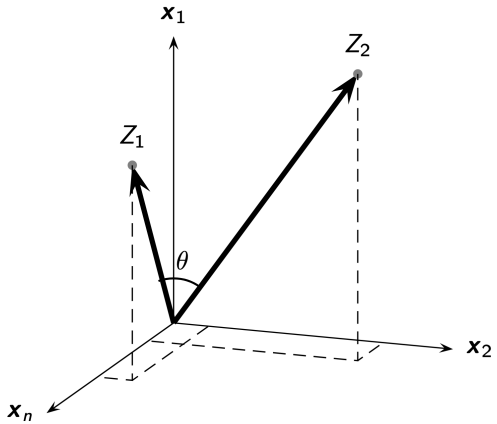
- Let $\overline{X}_1$ and $\overline{X}_2$ denote the centered attribute vectors in $\mathbb{R}^n$:

$$\overline{X}_1 = X_1 - \mathbf{1} \cdot \hat{\mu}_1 = \begin{pmatrix} x_{11} - \hat{\mu}_1 \\ x_{21} - \hat{\mu}_1 \\ \vdots \\ x_{n1} - \hat{\mu}_1 \end{pmatrix} \quad \overline{X}_2 = X_2 - \mathbf{1} \cdot \hat{\mu}_2 = \begin{pmatrix} x_{12} - \hat{\mu}_2 \\ x_{22} - \hat{\mu}_2 \\ \vdots \\ x_{n2} - \hat{\mu}_2 \end{pmatrix}$$

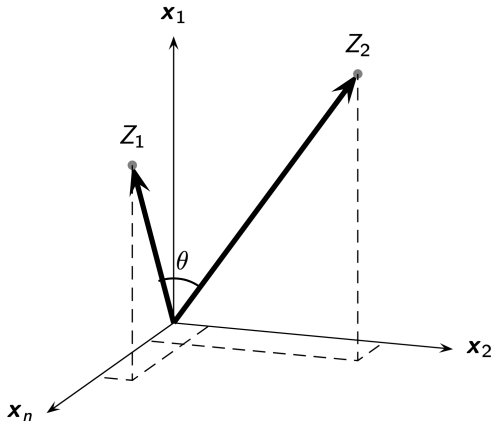
- Then, we have:

$$\hat{\sigma}_{12} = \frac{1}{n} \sum_{i=1}^{n} (x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2) = \frac{\overline{X}_1^T \overline{X}_2}{n}$$

That is, $\hat{\sigma}_{12}$ is the dot-product of the centered attribute vectors $\overline{X}_1$ and $\overline{X}_2$ normalized by the number of dimensions, n.

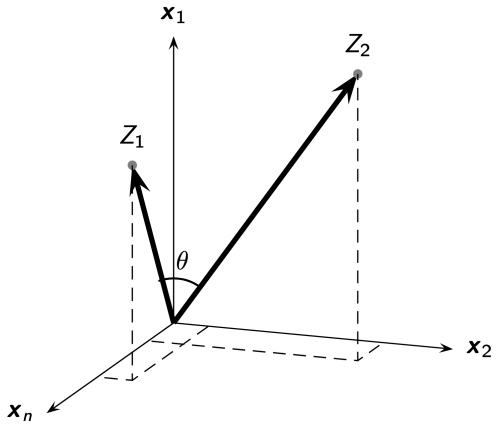


Geometric Interpretation of Sample Correlation:



Geometric Interpretation of Sample Correlation:

$$\hat{\rho}_{12} = \frac{\hat{\sigma}_{12}}{\hat{\sigma}_1 \hat{\sigma}_2} = \frac{\sum_{i=1}^{n}(x_{i1} - \hat{\mu}_1)(x_{i2} - \hat{\mu}_2)}{\sqrt{\sum_{i=1}^{n}(x_{i1} - \hat{\mu}_1)^2 \sum_{i=1}^{n}(x_{i2} - \hat{\mu}_2)^2}}$$



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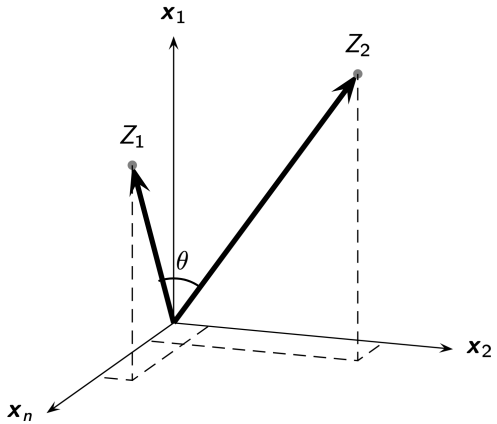
$$= \frac{\overline{X}_1^T \overline{X}_2}{\sqrt{\overline{X}_1^T \overline{X}_1}\sqrt{\overline{X}_2^T \overline{X}_2}} = \frac{\overline{X}_1^T \overline{X}_2}{\|\overline{X}_1\| \; \|\overline{X}_2\|} = \left(\frac{\overline{X}_1}{\|\overline{X}_1\|}\right)^T \left(\frac{\overline{X}_2}{\|\overline{X}_2\|}\right)$$



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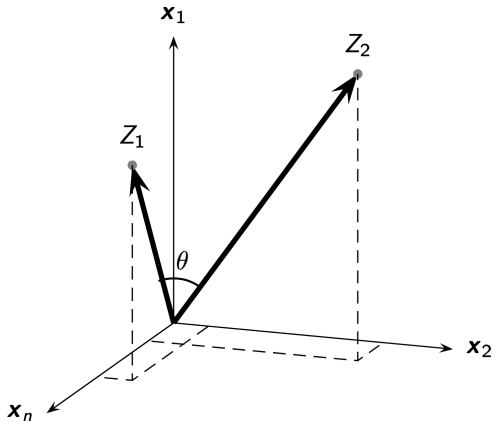
$$= \cos\theta$$



Geometric Interpretation of Sample Correlation:

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$$= \frac{\overline{X}_1^T \overline{X}_2}{\sqrt{\overline{X}_1^T \overline{X}_1}\sqrt{\overline{X}_2^T \overline{X}_2}} = \frac{\overline{X}_1^T \overline{X}_2}{\|\overline{X}_1\| \; \|\overline{X}_2\|} = \left(\frac{\overline{X}_1}{\|\overline{X}_1\|}\right)^T \left(\frac{\overline{X}_2}{\|\overline{X}_2\|}\right)$$

$$= \cos\theta$$

That is, $\hat{\rho}_{12}$ is the $\cos(\theta)$ where θ is the angle between the two centered attribute vectors $\overline{X}_1$ and $\overline{X}_2$.

Covariance Matrix

Definition

The variance–covariance information for the two attributes X_1 and X_2 can be summarized in the square 2×2 **covariance matrix** which is given by the following:

Covariance Matrix

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The variance–covariance information for the two attributes X_1 and X_2 can be summarized in the square 2×2 **covariance matrix** which is given by the following:

$$\boldsymbol{\Sigma} = E[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T]$$

Covariance Matrix

We can derive the following representation of the covariance matrix, based on its definition:

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Covariance Matrix

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$$
\begin{aligned}
\boldsymbol{\Sigma} &= E[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T] \\
&= E\left[\begin{pmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{pmatrix} \begin{pmatrix} X_1 - \mu_1 & X_2 - \mu_2 \end{pmatrix}\right] \\
&= E\left[\begin{pmatrix} (X_1 - \mu_1)^2 & (X_1 - \mu_1)(X_2 - \mu_2) \\ (X_2 - \mu_2)(X_1 - \mu_1) & (X_2 - \mu_2)^2 \end{pmatrix}\right]
\end{aligned}
$$

Covariance Matrix

We can derive the following representation of the covariance matrix, based on its definition:

$$
\begin{aligned}
\boldsymbol{\Sigma} &= E[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T] \\
&= E\left[\begin{pmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \end{pmatrix} \begin{pmatrix} X_1 - \mu_1 & X_2 - \mu_2 \end{pmatrix}\right] \\
&= E\left[\begin{pmatrix} (X_1 - \mu_1)^2 & (X_1 - \mu_1)(X_2 - \mu_2) \\ (X_2 - \mu_2)(X_1 - \mu_1) & (X_2 - \mu_2)^2 \end{pmatrix}\right] \\
&= \begin{pmatrix} E[(X_1 - \mu_1)^2] & E[(X_1 - \mu_1)(X_2 - \mu_2)] \\ E[(X_2 - \mu_2)(X_1 - \mu_1)] & E[(X_2 - \mu_2)^2] \end{pmatrix}
\end{aligned}
$$

Covariance Matrix

We can derive the following representation of the covariance matrix, based on its definition:

$$
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&= \begin{pmatrix} E[(X_1 - \mu_1)^2] & E[(X_1 - \mu_1)(X_2 - \mu_2)] \\ E[(X_2 - \mu_2)(X_1 - \mu_1)] & E[(X_2 - \mu_2)^2] \end{pmatrix} \\
\boldsymbol{\Sigma} &= \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix}
\end{aligned}
$$

Covariance Matrix

So, we have the following:

$$\boxed{\mathbf{\Sigma} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{21} & \sigma_2^2 \end{pmatrix}}$$

Due to the fact that $\sigma_{12} = \sigma_{21}$, $\mathbf{\Sigma}$ is *symmetric*.

Trace

Definition

The **trace** of $\boldsymbol{\Sigma}$ is the sum of the diagonal entries of $\boldsymbol{\Sigma}$, and is thus, equal to the total variance.

$$tr(\boldsymbol{\Sigma}) = var(\boldsymbol{D}) = \sigma_1^2 + \sigma_2^2$$

We know from this definition immediately that $tr(\boldsymbol{\Sigma}) \geq 0$.

Generalized Variance

Definition

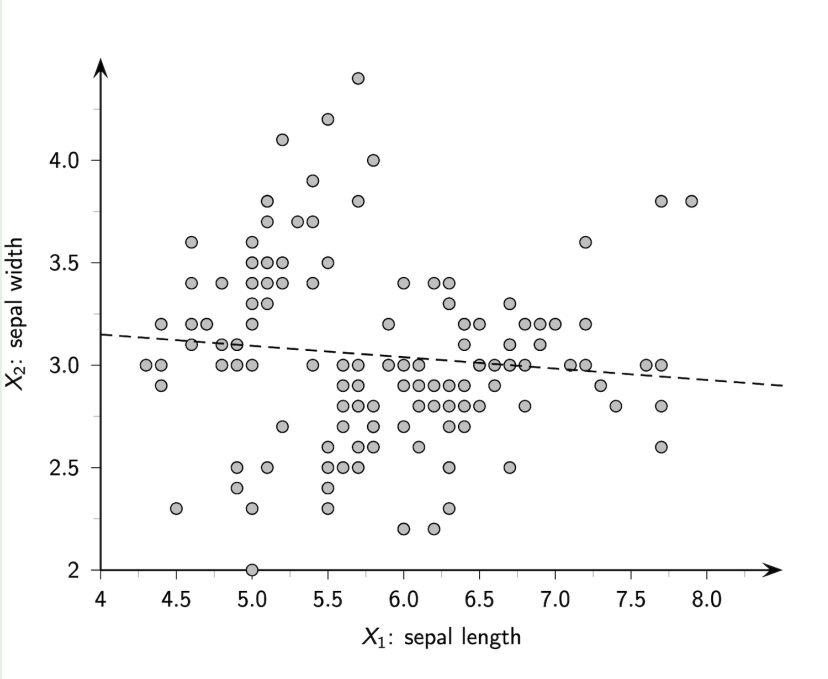
The **generalized variance** incorporates both the individual variances and the covariances as offsets to total variance, given as follows:

$$|\mathbf{\Sigma}| = \det(\mathbf{\Sigma}) = \sigma_1^2\sigma_2^2 - \sigma_{12}^2 = \sigma_1^2\sigma_2^2 - \rho_{12}^2\sigma_1^2\sigma_2^2 = (1-\rho_{12}^2)\sigma_1^2\sigma_2^2$$

Note that $|\rho_{12}| \leq 1$ implies that $\det(\mathbf{\Sigma}) \geq 0$.

Correlation: sepal length and sepal width

Example: Iris Dataset Extract: sepal length and sepal width



The sample mean is

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} 5.843 \\ 3.054 \end{pmatrix}$$

The sample covariance matrix is

$$\widehat{\boldsymbol{\Sigma}} = \begin{pmatrix} 0.681 & -0.039 \\ -0.039 & 0.187 \end{pmatrix}$$

The sample correlation is

$$\hat{\rho}_{12} = \frac{-0.039}{\sqrt{0.681 \cdot 0.187}} = -0.109$$

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Multivariate Analysis

Definition

Multivariate Analysis is the process of analyzing $d > 1$ attributes among objects in a population, typically using a statistical sample of n randomly selected objects.

Multivariate Analysis

$$\boldsymbol{D} = \begin{pmatrix} X_1 & X_2 & \cdots & X_d \\ \hline x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}$$

Multivariate Analysis

$$\boldsymbol{D} = \begin{pmatrix} X_1 & X_2 & \cdots & X_d \\ \hline x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}$$

Probabilistic (Row Vector) Perspective:

- Treat the data as a set of n points or vectors in the d-dimensional attribute space

$$\boldsymbol{x}_i = (x_{i1}, x_{i2}, \ldots, x_{id})^T \in \mathbb{R}^d$$

- $\boldsymbol{x}_i$'s constitute an independent, identically distributed (IID) randomly selected samples of n observations of the random variable $\boldsymbol{X} \in \mathbb{R}^d$

Multivariate Analysis

$$\boldsymbol{D} = \begin{pmatrix} X_1 & X_2 & \cdots & X_d \\ \hline x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}$$

Geometric (Column Vector) Perspective:

- Treat the sample as d vectors in n-dimensional space.

$$\boldsymbol{X}_j = (x_{1j}, x_{2j}, \ldots, x_{nj})^T \in \mathbb{R}^n$$

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Mean

In the probabilistic view, the d attributes are modeled as a vector random variable, $\boldsymbol{X} = (X_1, X_2, \ldots, X_d)^T$, and the points $\boldsymbol{x}_i$ are considered to be a random sample drawn from $\boldsymbol{X}$, i.e., IID with $\boldsymbol{X}$.

Definition

The **multivariate mean vector** is given by:

$$\boldsymbol{\mu} = E[\boldsymbol{X}] = \begin{pmatrix} \mu_1 & \mu_2 & \cdots & \mu_d \end{pmatrix}^T$$

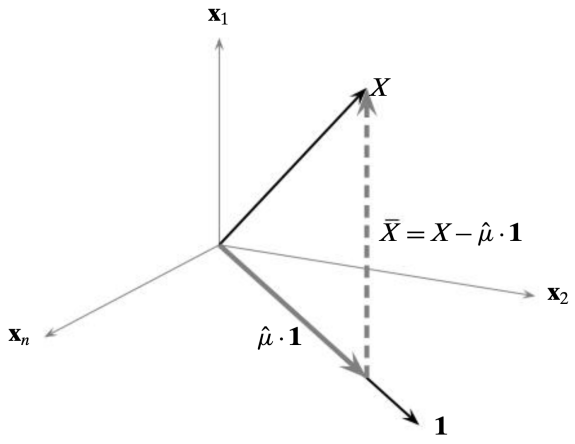
Sample Mean

Definition

The **sample mean** is given by the following:

$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{x}_i$$

Recall the following from the section on *univariate analysis*, for a single attribute vector, $X \in$



Geometric Perspective:

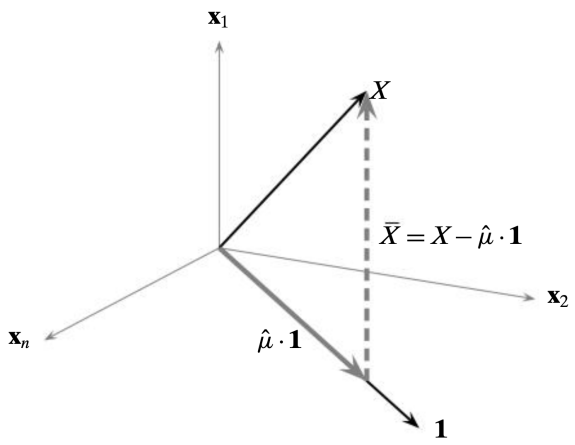
- Consider the projection of X onto the vector, $\mathbf{1}$.
- In $\mathbb{R}^n$, the projection, $\boldsymbol{p}$ of any vector, $\boldsymbol{b}$ onto vector $\boldsymbol{a}$ is given by:

$$\boldsymbol{p} = \left(\frac{\boldsymbol{a}^T \boldsymbol{b}}{\boldsymbol{a}^T \boldsymbol{a}} \right) \boldsymbol{a}$$

- In this case, we have:

$$\boldsymbol{p} = \left(\frac{X^T \mathbf{1}}{\mathbf{1}^T \mathbf{1}} \right) \mathbf{1} = \left(\frac{\sum_{i=1}^{n} x_i}{n} \right) \mathbf{1} = \hat{\mu} \cdot \mathbf{1}$$

Geometric Interpretation of Multivariate Sample Mean



Geometric Perspective:

- Use the same idea in the multivariate case, each of the d attribute vectors projected onto the ones vector, $\mathbf{1} \in \mathbb{R}^n$.
- Multivariate mean vector can be expressed as:

$$\hat{\boldsymbol{\mu}} = \frac{1}{n} \boldsymbol{D}^T \mathbf{1}$$

Covariance Matrix

Definition

The **covariance matrix** is a $d \times d$ (square) symmetric matrix

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1d} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2d} \\ \cdots & \cdots & \cdots & \cdots \\ \sigma_{d1} & \sigma_{d2} & \cdots & \sigma_d^2 \end{pmatrix}$$

Covariance Matrix is Positive Semidefinite

$\boldsymbol{\Sigma}$ is a *positive semidefinite* matrix, that is,

$$\boldsymbol{a}^T \boldsymbol{\Sigma} \boldsymbol{a} \geq 0 \text{ for any } d\text{-dimensional vector } \boldsymbol{a}$$

To see this, observe that

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$$\begin{aligned} \boldsymbol{a}^T \boldsymbol{\Sigma} \boldsymbol{a} &= \boldsymbol{a}^T E\big[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T\big] \boldsymbol{a} \\ &= E\big[\boldsymbol{a}^T (\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T \boldsymbol{a}\big] \end{aligned}$$

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\boldsymbol{a}^T \boldsymbol{\Sigma} \boldsymbol{a} &= \boldsymbol{a}^T E\big[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T\big] \boldsymbol{a} \\
&= E\big[\boldsymbol{a}^T (\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^T \boldsymbol{a}\big] \\
&= E\big[Y^2\big] \qquad \text{where } Y = \boldsymbol{a}^T(\boldsymbol{X} - \boldsymbol{\mu}) = \sum_{i=1}^{d} a_i (X_i - \mu_i)
\end{aligned}$$

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&\geq 0 \quad \text{since the expectation of a squared RV is non-negative.}
\end{aligned}$$

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\end{aligned}$$

Because $\boldsymbol{\Sigma}$ is also symmetric, this implies that all the eigenvalues of $\boldsymbol{\Sigma}$ are real and non-negative, and they can be arranged from the largest to the smallest as follows:

Covariance Matrix is Positive Semidefinite

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&\geq 0 \quad \text{since the expectation of a squared RV is non-negative.}
\end{aligned}$$

Because $\boldsymbol{\Sigma}$ is also symmetric, this implies that all the eigenvalues of $\boldsymbol{\Sigma}$ are real and non-negative, and they can be arranged from the largest to the smallest as follows: $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d \geq 0$.

Total Variance

Definition

The total variance is given as:

$$var(\boldsymbol{D}) = tr(\boldsymbol{\Sigma}) = \sum_{i=1}^{d} \sigma_i^2$$

Generalized Variance

Definition

The generalized variance is defined as the determinant of the covariance matrix, $\det(\mathbf{\Sigma})$, also denoted $|\mathbf{\Sigma}|$:

$$\det(\mathbf{\Sigma}) = \prod_{i=1}^{d} \lambda_i \geq 0$$

Since the eigenvalues of $\mathbf{\Sigma}$ are non-negative ($\lambda_i \geq 0$), it follows that $\det(\mathbf{\Sigma}) \geq 0$.

Sample Covariance Matrix

Definition

The **sample covariance matrix** is a $d \times d$ (square) symmetric matrix

$$\widehat{\mathbf{\Sigma}} = \begin{pmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} & \cdots & \hat{\sigma}_{1d} \\ \hat{\sigma}_{21} & \hat{\sigma}_2^2 & \cdots & \hat{\sigma}_{2d} \\ \cdots & \cdots & \cdots & \cdots \\ \hat{\sigma}_{d1} & \hat{\sigma}_{d2} & \cdots & \hat{\sigma}_d^2 \end{pmatrix}$$

Geometric Perspective of Sample Covariance Matrix

Let $\overline{\boldsymbol{D}}$ represent the centered data matrix

$$\overline{\boldsymbol{D}} = \boldsymbol{D} - \mathbf{1} \cdot \hat{\boldsymbol{\mu}}^T = \begin{pmatrix} \boldsymbol{x}_1^T - \hat{\boldsymbol{\mu}}^T \\ \boldsymbol{x}_2^T - \hat{\boldsymbol{\mu}}^T \\ \vdots \\ \boldsymbol{x}_n^T - \hat{\boldsymbol{\mu}}^T \end{pmatrix} = \begin{pmatrix} - & \overline{\boldsymbol{x}}_1^T & - \\ - & \overline{\boldsymbol{x}}_2^T & - \\ & \vdots & \\ - & \overline{\boldsymbol{x}}_n^T & - \end{pmatrix}$$

Inner product and outer product form for sample covariance matrix:

$$\widehat{\boldsymbol{\Sigma}} = \frac{1}{n}\left(\overline{\boldsymbol{D}}^T\ \overline{\boldsymbol{D}}\right) = \frac{1}{n}\begin{pmatrix} \overline{X}_1^T\,\overline{X}_1 & \overline{X}_1^T\,\overline{X}_2 & \cdots & \overline{X}_1^T\,\overline{X}_d \\ \overline{X}_2^T\,\overline{X}_1 & \overline{X}_2^T\,\overline{X}_2 & \cdots & \overline{X}_2^T\,\overline{X}_d \\ \vdots & \vdots & \ddots & \vdots \\ \overline{X}_d^T\,\overline{X}_1 & \overline{X}_d^T\,\overline{X}_2 & \cdots & \overline{X}_d^T\,\overline{X}_d \end{pmatrix} \quad \widehat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum_{i=1}^{n} \overline{\boldsymbol{x}}_i \cdot \overline{\boldsymbol{x}}_i^T$$

i.e., $\widehat{\boldsymbol{\Sigma}}$ is given as the pairwise *inner or dot products* of the centered attribute vectors, normalized by the sample size, or as a sum of rank-one matrices obtained as the *outer product* of each centered point.

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Range Normalization

If the attribute values are in vastly different scales, then it is necessary to normalize them.

Definition

Range Normalization: Let X be an attribute and let $x_1, x_2, \ldots, x_n$ be a random sample drawn from X. In *range normalization* each value is scaled by the sample range $\hat{r}$ of X:

$$x_i' = \frac{x_i - \min_i\{x_i\}}{\hat{r}} = \frac{x_i - \min_i\{x_i\}}{\max_i\{x_i\} - \min_i\{x_i\}}$$

After transformation the new attribute takes on values in the range $[0, 1]$.

Standard Score Normalization

Definition

Standard Score Normalization: Also called z-normalization; each value is replaced by its z-score:

$$x_i' = \frac{x_i - \hat{\mu}}{\hat{\sigma}}$$

where $\hat{\mu}$ is the sample mean and $\hat{\sigma}^2$ is the sample variance of X. After transformation, the new attribute has mean $\hat{\mu}' = 0$, and standard deviation $\hat{\sigma}' = 1$.

Normalization Example

$\boldsymbol{x}_i$	Age (X_1)	Income (X_2)
$\boldsymbol{x}_1$	12	300
$\boldsymbol{x}_2$	14	500
$\boldsymbol{x}_3$	18	1000
$\boldsymbol{x}_4$	23	2000
$\boldsymbol{x}_5$	27	3500
$\boldsymbol{x}_6$	28	4000
$\boldsymbol{x}_7$	34	4300
$\boldsymbol{x}_8$	37	6000
$\boldsymbol{x}_9$	39	2500
$\boldsymbol{x}_{10}$	40	2700

Since `Income` is much larger, it dominates `Age`. The sample range for `Age` is $\hat{r} = 40 - 12 = 28$, whereas for `Income` it is $6000 - 300 = 5700$. For range normalization, the point $\boldsymbol{x}_2 = (14, 500)$ is scaled to

$$\boldsymbol{x}_2' = \left(\frac{14 - 12}{28}, \frac{500 - 300}{5700}\right) = (0.071, 0.035)$$

For z-normalization, we have

	Age	Income
$\hat{\mu}$	27.2	2680
$\hat{\sigma}$	9.77	1726.15

Thus, $\boldsymbol{x}_2 = (14, 500)$ is scaled to

$$\boldsymbol{x}_2' = \left(\frac{14 - 27.2}{9.77}, \frac{500 - 2680}{1726.15}\right) = (-1.35, -1.26)$$

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Univariate Normal Distribution

The normal distribution plays an important role as the parametric distribution of choice in clustering, density estimation, and classification.

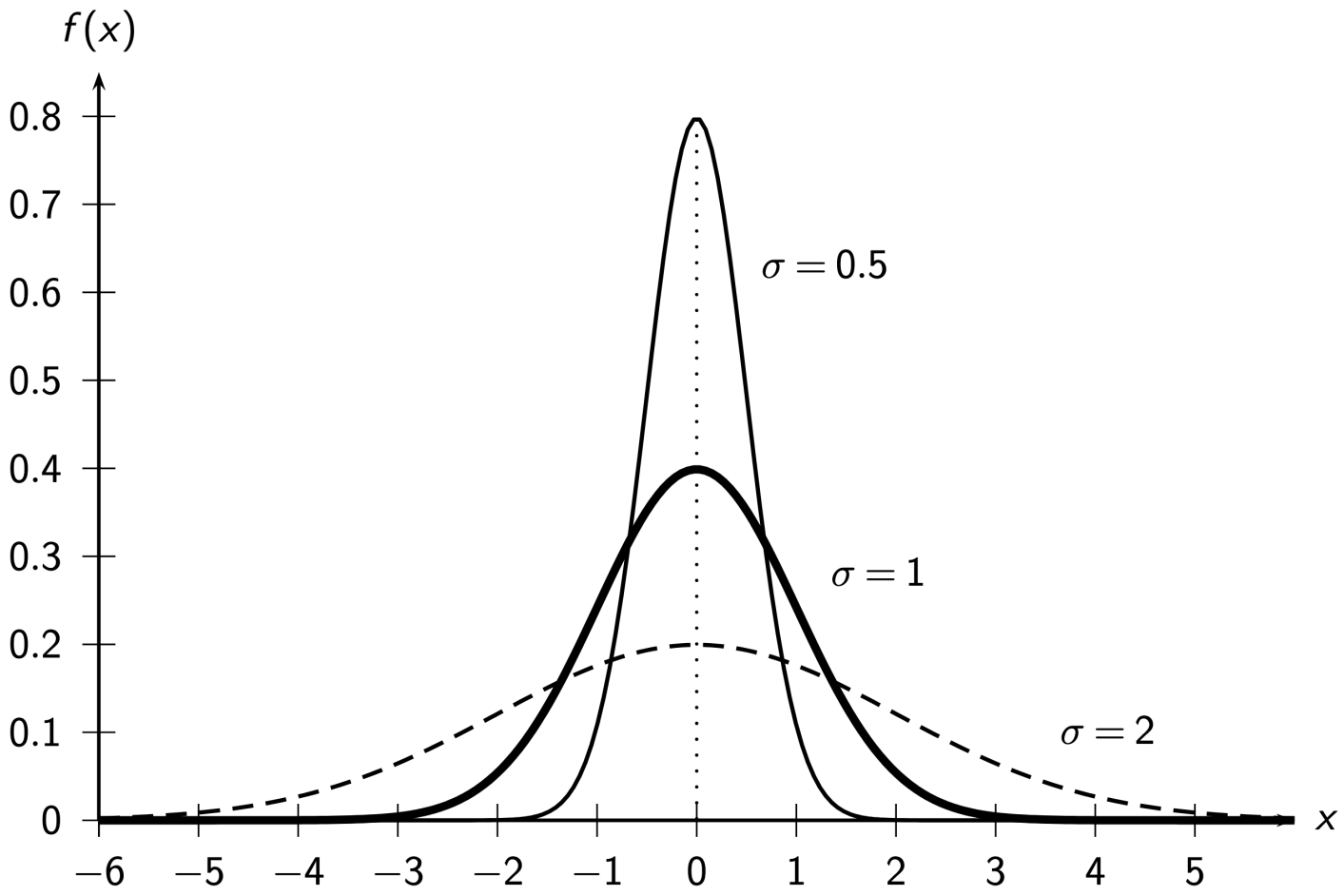
Definition

A random variable X has a normal distribution, with the parameters mean μ and variance σ^2, if the probability density function of X is given as follows:

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \quad \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

- The $(x-\mu)^2$ term measures the distance of a value x from the mean μ of the distribution, and thus the probability density decreases exponentially as a function of the distance from the mean.
- The maximum value of the density occurs at the mean value $x = \mu$, given as $f(\mu) = \frac{1}{\sqrt{2\pi\sigma^2}}$, which is inversely proportional to the standard deviation σ of the distribution.

Univariate Normal Distribution



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Multivariate Normal Distribution

Definition

Given the d-dimensional vector random variable $\boldsymbol{X} = (X_1, X_2, \ldots, X_d)^T$, it has a multivariate normal distribution, with the parameters mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, if its joint multivariate probability density function is given as follows:

$$f(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(\sqrt{2\pi})^d \sqrt{|\boldsymbol{\Sigma}|}} \exp\left\{-\frac{(\boldsymbol{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}-\boldsymbol{\mu})}{2}\right\}$$

where $|\boldsymbol{\Sigma}|$ is the determinant of the covariance matrix.

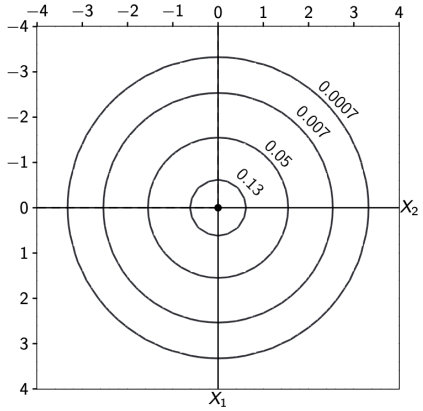
- The term

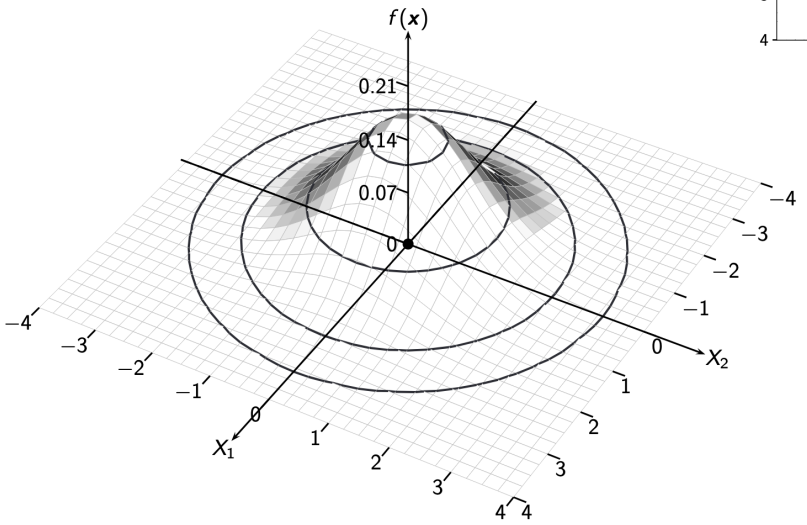
$$(\boldsymbol{x}_i - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}_i - \boldsymbol{\mu})$$

measures the distance, called the *Mahalanobis distance* of the point $\boldsymbol{x}$ from the mean $\boldsymbol{\mu}$ of the distribution, taking into account all of the variance–covariance information between the attributes.

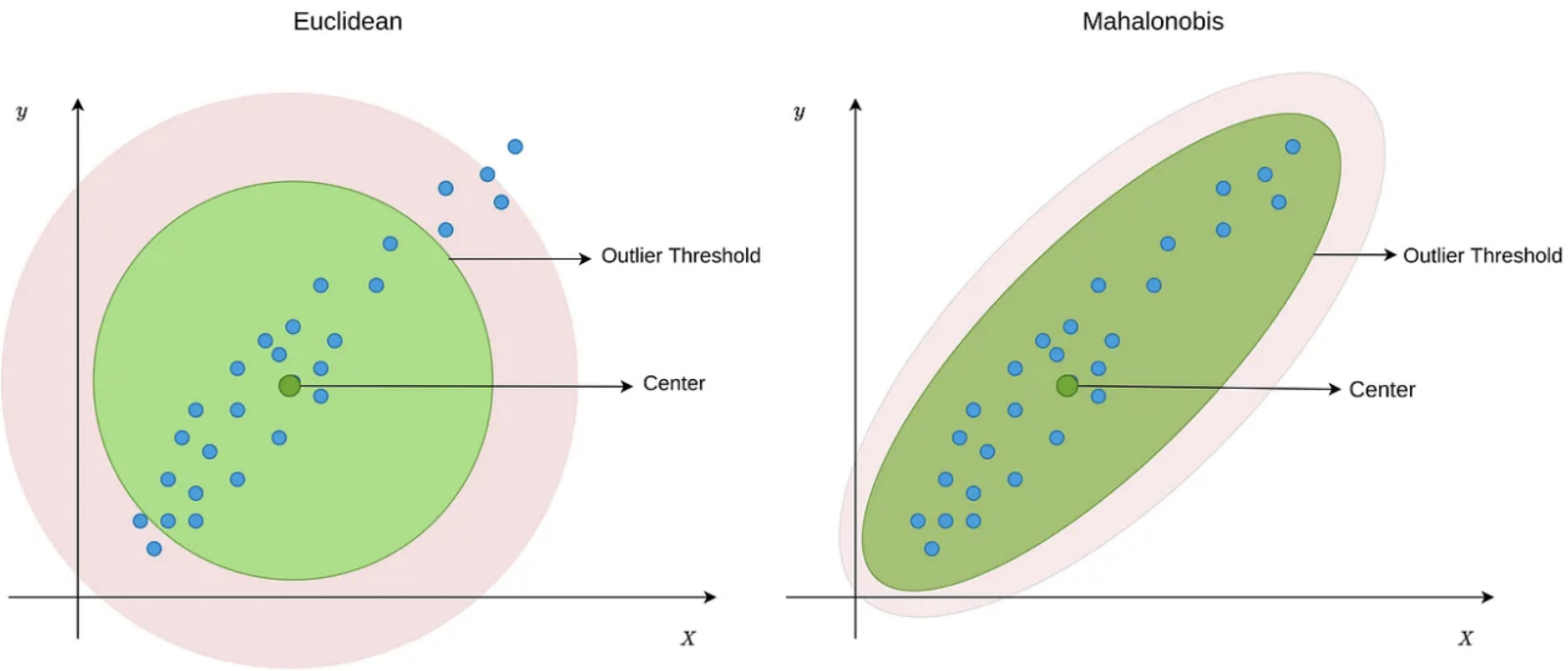
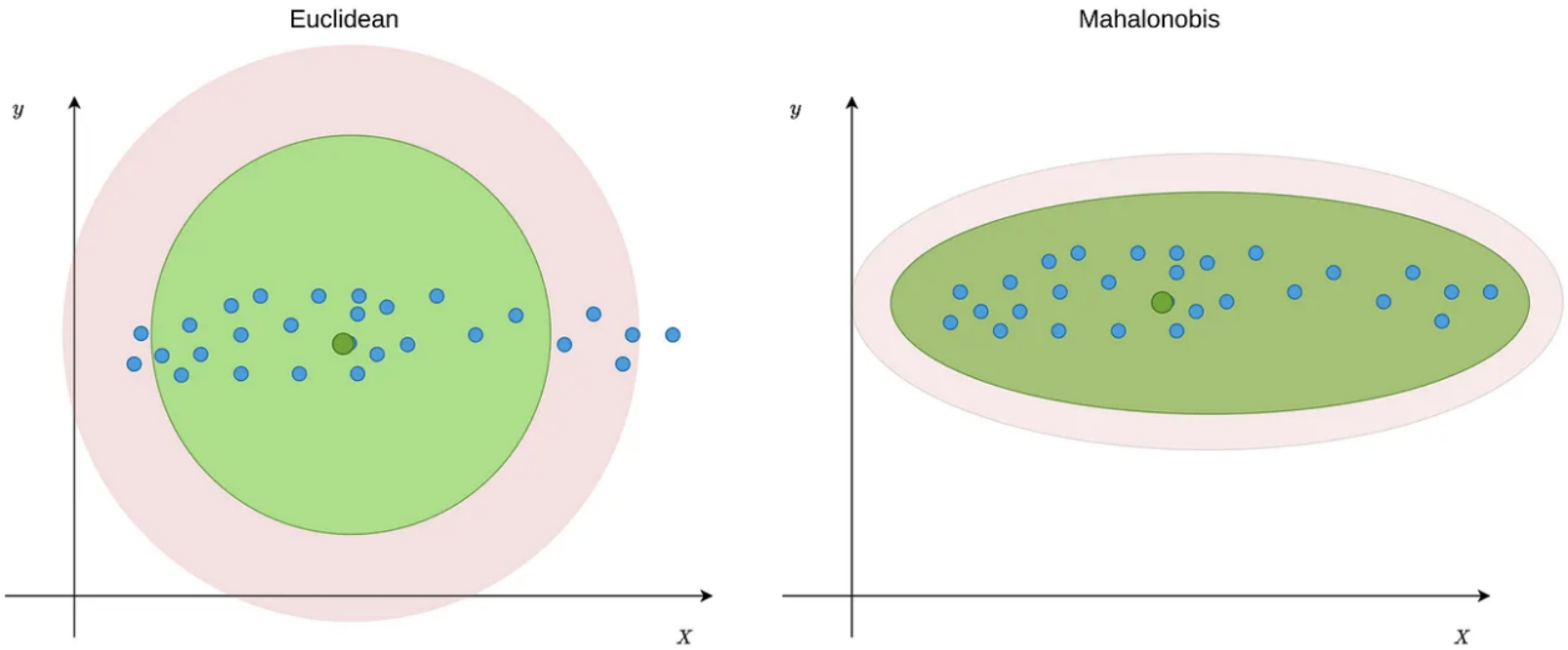
Multivariate Normal Distribution

Parameters: $\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\Sigma = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$





Multivariate Normal Distribution



Multivariate Normal Distribution